

CUSTOMER CHURN INTELLIGENCE REPORT

An end-to-end churn analysis built on customer, subscription, and support data — covering data cleaning, feature engineering, KPI computation, and visualization to identify why customers leave and where revenue is most at risk.



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Tools & Technologies: Python • SQLite • Pandas • NumPy • Matplotlib • Seaborn

Executive Summary

This report analyzes customer churn for a subscription-based business using customer demographic, subscription, and support-interaction data. Raw data was consolidated into a SQLite database, cleaned, and merged into a single analytical dataset, from which churn, retention, and revenue-risk metrics were derived. The analysis finds an overall churn rate of **28.57%**, with churn concentrated heavily among **Basic plan** subscribers (60% churn) and customers who raised escalated support tickets — escalations correlate strongly with churn at **0.77**. Approximately **Rs. 73.94K** in monthly recurring revenue is tied to already-churned customers, underscoring the financial stakes of retention efforts.

Metric	Value	Interpretation
Churn Rate	28.57%	Roughly 1 in 3.5 customers has churned
Retention Rate	71.43%	Majority of customers remain active
Avg. Revenue Per User (ARPU)	Rs. 18.85	Baseline monthly value per customer
Avg. Customer Tenure	1,539 days (~4.2 yrs)	Long average lifetime, skewed by loyal customers
Revenue at Risk	Rs. 73.94K	Monthly revenue lost from churned accounts
Escalation Rate	19.05%	Share of customers with an escalated complaint
Avg. Complaints per User	0.43	Most customers file 0-1 complaints
Escalation ↔ Churn Correlation	0.77	Strong positive relationship

Top takeaway: Retention efforts should prioritize two levers — migrating or incentivizing Basic-plan customers toward higher tiers, and resolving support escalations faster, since escalated tickets are the single strongest predictor of churn observed in this dataset.

1. Project Overview & Methodology

The analysis followed a standard end-to-end data analytics workflow, moving from raw, disparate data sources to a clean, unified dataset ready for KPI computation and visualization.

- 1. Data Ingestion** — Raw customer, subscription, and support data (originally in Excel) were loaded into a SQLite database, with each sheet converted into its own relational table.
- 2. Data Cleaning** — Column names were standardized, irrelevant fields dropped, date fields converted to proper datetime types, categorical values normalized (e.g. Men/Women → Male/Female), and missing country values imputed using state-to-country mapping.
- 3. Feature Engineering** — New analytical fields were derived: a binary churn flag, customer tenure in days, per-customer complaint counts, and a three-tier churn risk classification (low / medium / high).
- 4. Table Merging** — Customer, subscription, and support tables were joined on customer ID, with careful shape validation at each join to avoid row duplication from one-to-many relationships.
- 5. KPI Computation** — Core business metrics — churn rate, retention rate, ARPU, tenure, revenue at risk, escalation rate, and complaint frequency — were computed directly from the merged dataset.
- 6. Visualization & Correlation** — Trends and relationships were visualized using Matplotlib and Seaborn, including a correlation heatmap built from properly ordinal-encoded categorical features.

Data Sources

Table	Description
customer	Demographics: customer name, gender, date of birth, state, country
subscription	Plan type, contract type, monthly charges, start/renewal/cancellation dates, churn score
support	Support complaints, escalation flags, complaint dates

2. Visual Insights

2.1 Monthly Churn Trend

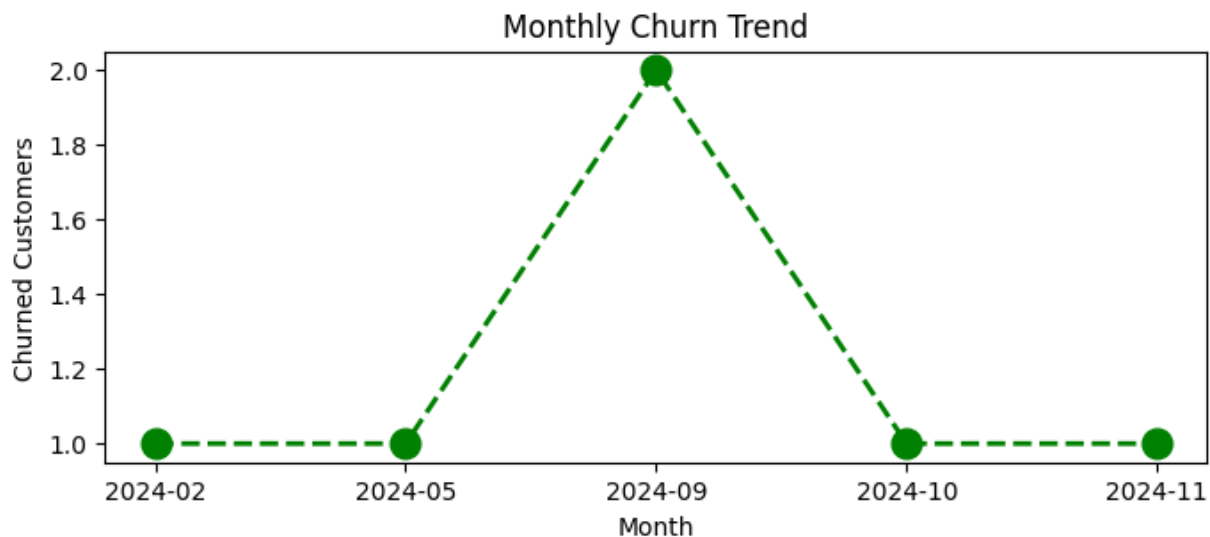


Figure 1: Number of customers churned per month, based on cancellation date.

Churn events are tracked monthly by grouping cancellation dates. The trend line highlights whether churn is accelerating, stable, or seasonal — a rising trend would signal an emerging retention problem, while spikes in specific months can be cross-checked against pricing changes, service incidents, or competitor activity around the same period.

2.2 Churn Rate by Plan Type

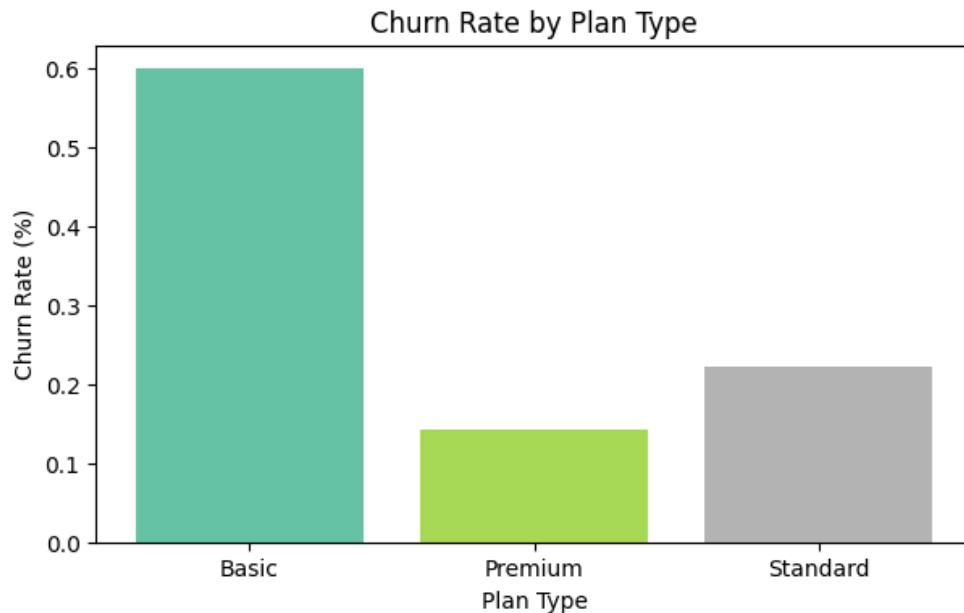


Figure 2: Churn rate segmented by subscription plan (Basic, Standard, Premium).

This is the most actionable chart in the analysis. **Basic plan** subscribers churn at **60%**, more than double the Standard tier (22.22%) and over 4x the Premium tier (14.29%). This strongly suggests that Basic-tier customers perceive less value or stickiness in the product, making plan-tier migration and Basic-tier engagement campaigns high-priority retention levers.

2.3 Churn Rate by State

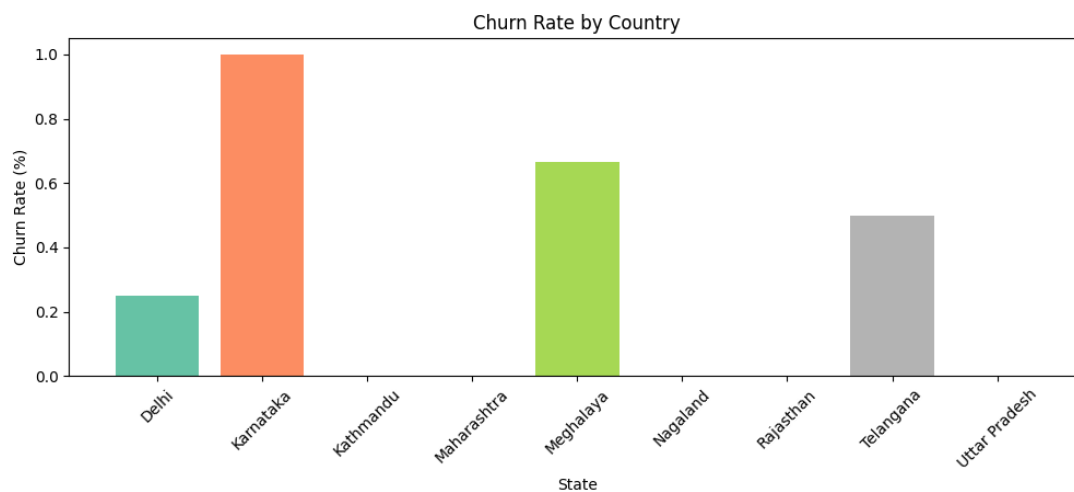


Figure 3: Geographic distribution of churn rate across states.

Churn is not uniform across geographies. States with notably higher churn rates may indicate regional service quality gaps, weaker local support coverage, or stronger regional competition, and warrant further investigation at the market level.

2.4 Correlation Heatmap

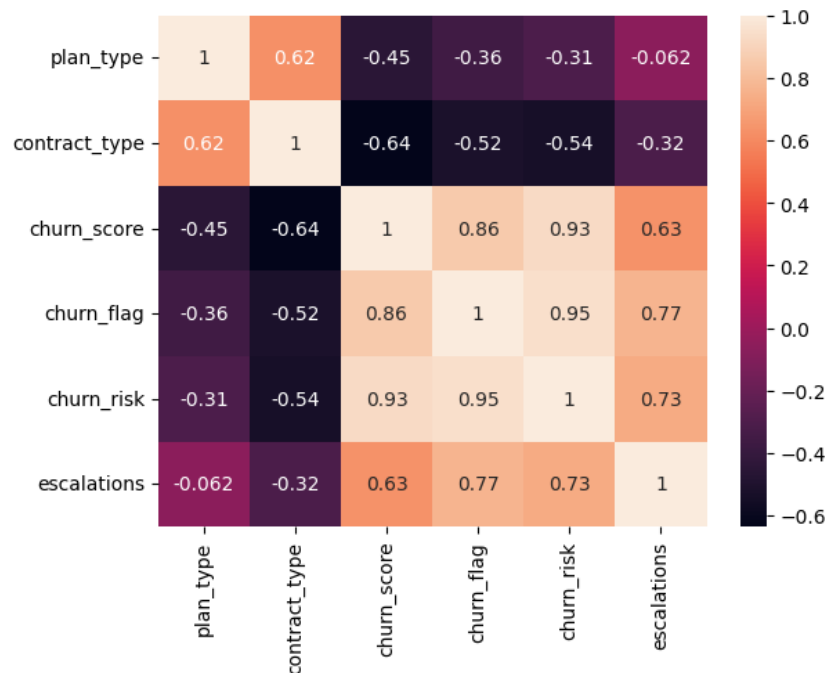


Figure 4: Correlation matrix of ordinally-encoded plan type, contract type, churn score, churn flag, churn risk, and escalations.

Categorical features (plan type, contract type, churn risk) were ordinally encoded — preserving the natural ranking (e.g. Basic < Standard < Premium) rather than arbitrary category codes — before computing correlations, since naive encoding would produce misleading relationships. The heatmap confirms that **churn score** and **escalations** are the features most strongly associated with churn, while plan type shows a meaningful negative relationship with churn (higher-tier plans, lower churn).

2.5 Plan Type vs. Monthly Charges by Risk Segment

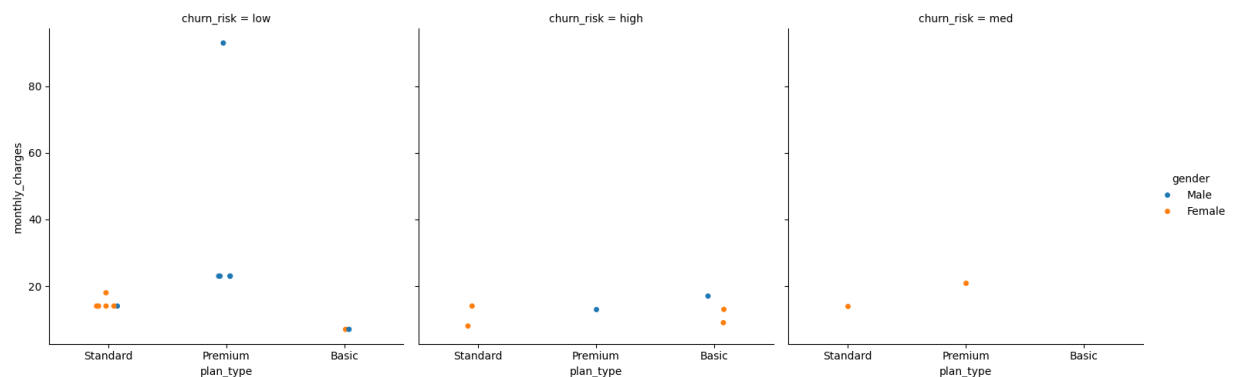


Figure 5: Monthly charges by plan type and gender, faceted by churn risk (low / medium / high).

Faceting by churn risk segment shows how spend and plan mix shift as risk increases, helping identify whether high-risk customers are concentrated in specific plan and pricing combinations that may need

targeted intervention.

3. Key Findings

- **Escalations are the strongest churn signal observed** (correlation = 0.77). Customers who escalate a support issue are far more likely to churn than those who don't, making fast, effective escalation resolution one of the highest-leverage retention interventions available.
- **Basic-plan customers are the highest churn-risk segment**, churning at 60% versus 22.22% (Standard) and 14.29% (Premium). This 4x gap between Basic and Premium suggests the entry-tier plan may be under-delivering on perceived value.
- **Revenue at risk is concentrated, not distributed**: Rs. 73.94K in monthly charges sits with customers who have already churned — a recoverable pool if win-back campaigns are targeted correctly.
- **Escalation rate (19.05%) is meaningfully lower than complaint volume would suggest** (avg. 0.43 complaints/user), meaning most complaints are resolved without escalating — but the ones that do escalate carry outsized churn risk.
- **Average tenure of ~1,539 days (~4.2 years)** indicates a generally loyal customer base overall, so churn is likely driven by a specific subset of dissatisfied customers rather than broad, systemic attrition.
- **Geographic variation in churn** suggests regional factors (support coverage, local competition) play a role and merit state-level follow-up analysis.

Recommendations

- Prioritize and fast-track escalated support tickets given their outsized correlation with churn.
- Design a Basic-to-Standard upgrade incentive (e.g. limited-time discount or added feature) to reduce the Basic-tier churn gap.
- Launch a win-back campaign targeted at the Rs. 73.94K in at-risk/already-churned revenue, prioritizing customers with short tenure and no prior escalation.
- Investigate high-churn states individually to determine whether the driver is support quality, pricing perception, or competitive pressure.

4. Conclusion

This project demonstrates a complete churn analytics pipeline — from raw multi-source data through cleaning, feature engineering, and merging, to KPI derivation and visual analysis. The findings paint a clear picture: churn in this customer base is not random, but concentrated among Basic-plan subscribers and customers whose support issues escalate. With a 28.57% churn rate and Rs. 73.94K in at-risk monthly revenue, targeted interventions on these two fronts — plan-tier engagement and escalation handling — represent the clearest path to measurably improving retention.

Tech Stack

Python (NumPy, Pandas) • SQLite (sqlite3) • Matplotlib • Seaborn • Jupyter Notebook

About the Author



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